



DEMO1_w2

SMART ENERGY ARBITRAGE

Decision-aware BESS arbitrage demo: what is ready now?

Can forecast-to-optimize improve Ukrainian battery arbitrage under real DAM prices?

CONTROL DEFAULT

strict_similar_da

best stability / median regret

PROMISING RESEARCH

horizon-aware TFT

lowest mean regret diagnostic

LIVE SURFACE

Operator +
Defense

FastAPI / Postgres / Dagster demo

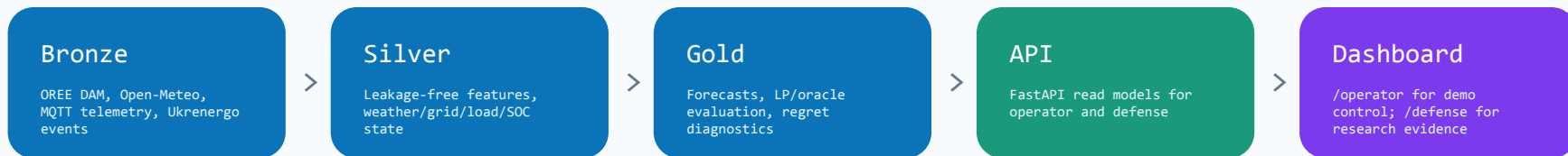
CLAIM BOUNDARY

DT not live

policy preview is not materialized

- Observed OREE DAM + tenant weather feed the benchmark, not random split data.
- Every strategy is scored through the same LP/oracle regret evaluator and Level 1 battery model.
- Dashboard separates production-ready control from research primitives.

Architecture: medallion data pipeline to operator decision surface



- Dagster asset groups show lineage from Bronze to Silver to Gold.
- Postgres stores benchmark rows, telemetry/read models, and research diagnostics.
- MLflow tracks benchmark and research runs; official NBEATSx/TFT adapters are present but smoke-scale.

Main benchmark result: the simple control is still hard to beat



- Strict similar-day has the best median regret and strongest stability, so it remains the default comparator.
- Raw compact TFT and NBEATSx are academically useful candidates, but not operational winners.
- This benchmark rejects weaker ML candidates instead of claiming automatic ML superiority.

Horizon-aware TFT is the strongest research signal, not yet production control



BEST MEAN REGRET

834 UAH

horizon-aware TFT diagnostic

WHY NOT DEFAULT

median/wins

strict is still more stable

THESIS CLAIM

bridge to DFL

not full differentiable training

Horizon-aware means the correction is learned per forecast horizon hour, preserving the 24-hour price shape the LP optimizer actually uses.

DFL/DT status: evidence exists, live policy does not

Ready

DFL training examples

Rows with features, LP action, oracle value, regret, degradation and throughput are persisted.

Pilot

Regret-weighted calibration

Weighted MAE improved for one tenant, but strict LP re-evaluation is neutral/negative.

🙄 Not live

Decision Transformer

Policy preview correctly reports not materialized; no market execution claim.

Academic boundary: DFL and DT are framed as research primitives until strict evaluation proves feasibility and value.

What to show in the live demo

1 **Dagster asset groups** Show Bronze/Silver/Gold lineage and thesis-grade pipeline discipline.

2 **/operator** Show live SOC, baseline LP preview, forecast stack panel, gatekeeper/readiness, schedule dock.

3 **/defense** Show benchmark table, claim boundaries, diagnostics, DFL/DT readiness.

4 **MLflow** Show experiment lineage; explain benchmark/readiness runs.

OPERATOR URL
localhost:64164/operator

main product demo

DEFENSE URL
localhost:64164/defense

research evidence

DAGSTER
localhost:3001

lineage /
materializations

MLFLOW
localhost:5000

experiments

Demo talk track: strongest honest conclusion

“We built a reproducible forecast-to-optimize research system for Ukrainian BESS arbitrage. The strict similar-day control remains the stable production comparator; horizon-aware TFT is the strongest research signal and motivates full DFL next.”

- Do claim: observed-data benchmark, leakage-free rolling origin, LP/oracle regret scoring, telemetry/SOC-aware operator read model.
- Do claim: horizon-aware TFT diagnostic beats strict on mean regret but is not stable enough to replace strict control.
- Do not claim: full digital twin, live trading, full SOTA training, or deployed Decision Transformer policy.

Next after demo: **materialize official NBEATSx/TFT live inference, train/evaluate DT offline, then promote only strategies that beat strict under strict simulator checks.**